Tutorial on St. Venant principals\

Problem 1

```
% Equilateral triangular cross section
clear;
% Prandtl's stress function :

syms x y phi G K a
phi = G*K*(0.5*(x^2+y^2)-(x^3-3*x*y^2)/(2*a)-2*a^2/27)
```

```
phi =
```

$$G\,K\left(\frac{3\,x\,y^2-x^3}{2\,a}-\frac{2\,a^2}{27}+\frac{x^2}{2}+\frac{y^2}{2}\right)$$

```
t_xz = diff(phi,y)
```

```
t_xz =
```

$$G\,K\left(y+\frac{3\,x\,y}{a}\right)$$

```
t_yz = diff(phi,x)
```

```
t_yz =
```

$$G\,K\left(x-\frac{3\,x^2-3\,y^2}{2\,a}\right)$$

Defining the zone

```
a=1; G=1; K=1;
N = 501;

x = linspace(-a/3,2*a/3,N);
y = linspace(-a/sqrt(3),a/sqrt(3),N);

[X,Y] = meshgrid(x,y);


inside = (X >= -a/3) & ...
    (Y <= (2*a/3-X)/sqrt(3)) & ...
    (Y >= -(2*a/3-X)/sqrt(3));

tau_xz = -G*K*(Y+3*X.*Y/a);  tau_yz= G*K*(X-(3.*X.^2-3.*Y.^2)/(2*a));
% actually it is the Phi/(GK) though I haver chosen G=K=1
Phi = -G*K*(0.5*(X.^2 + Y.^2) - (X.^3 - 3*X.*Y.^2)/(2*a) - 2*a^2/27);
```

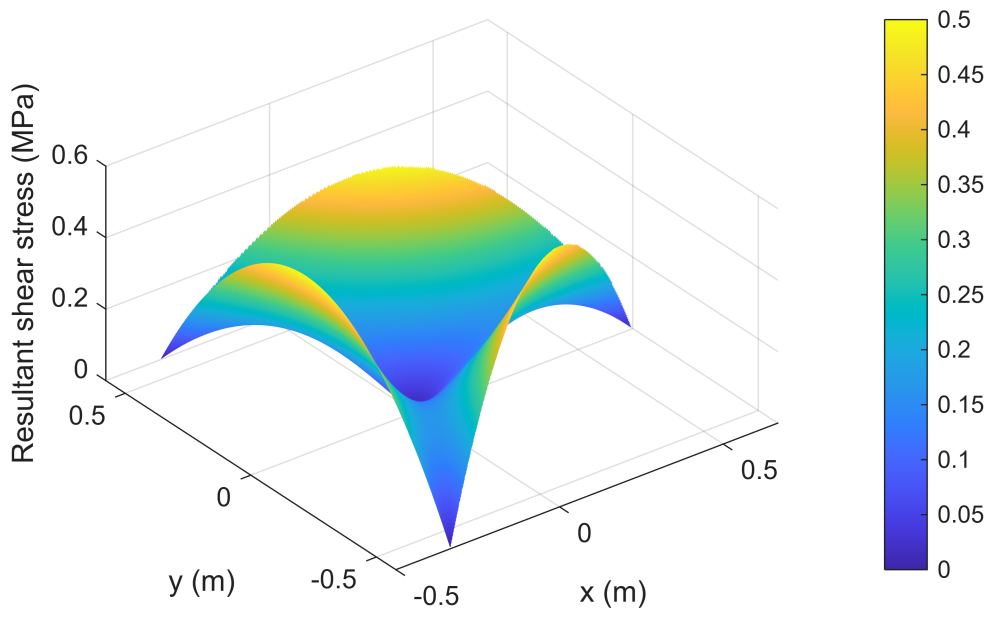
```
tau_xz(~inside) = NaN;
tau_yz(~inside) = NaN;
tau = sqrt(tau_xz.^2 + tau_yz.^2);
Phi(~inside) = NaN;

% PLOT PRANDTL STRESS FUNCTION
figure('Color','w');

surf(X,Y,Phi,'EdgeColor','none');
daspect([1 1 0.2]); colorbar;
xlabel('x (m)');
ylabel('y (m)');
zlabel('Stress Function Distribution');
% 3D shear stress resultant5 plot
figure('Color','w');

surf(X, Y, tau, tau, ...
    'EdgeColor','none');

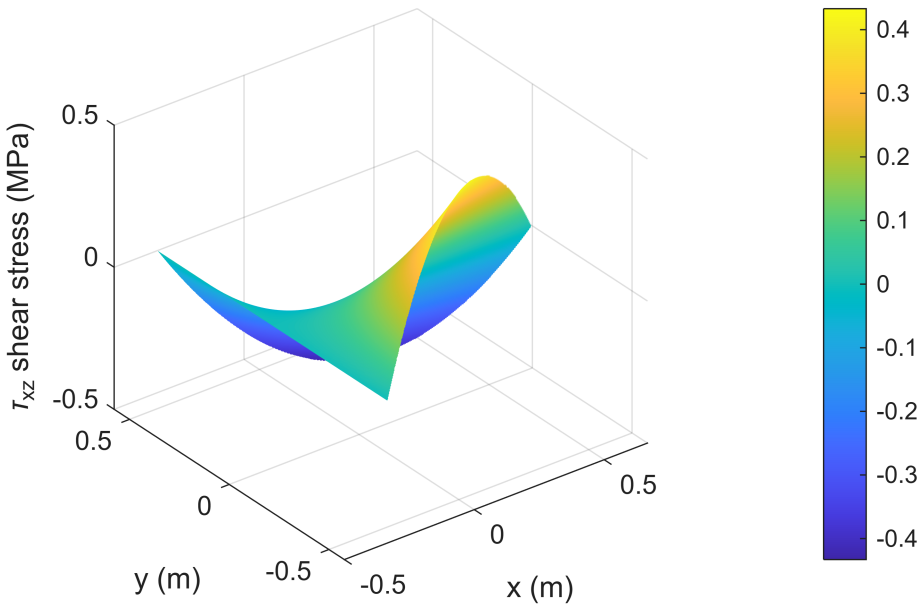
daspect([1 1 1]);
colorbar;
xlabel('x (m)');
ylabel('y (m)');
zlabel('Resultant shear stress (MPa)');
```




```
surf(X, Y, tau_xz, tau_xz, ...
    'EdgeColor','none');

daspect([1 1 1]);
colorbar;
xlabel('x (m)');
```

```
ylabel('y (m)');
zlabel('\tau_{xz} shear stress (MPa)');
```




```
surf(X, Y, tau_yz, tau_yz, ...
    'EdgeColor','none');

daspect([1 1 1]);
colorbar;
xlabel('x (m)');
ylabel('y (m)');
zlabel('\tau_{yz} shear stress (MPa)');
% PLOT SHEAR-STRESS VECTORS
% Subsample the grid to keep the arrows readable

figure('Color','w');

skip = 25;
Xs = X(1:skip:end,1:skip:end);
Ys = Y(1:skip:end,1:skip:end);

Tx = tau_xz(1:skip:end,1:skip:end);
Ty = tau_yz(1:skip:end,1:skip:end);

quiver(Xs,Ys,Tx,Ty,'k');
hold on;


axis equal;
grid on;
xlabel('x (m)');
```

```
ylabel('y (m)');
title('Shear-Stress Component Field');


% PLOT RESULTANT SHEAR-STRESS CONTOUR
figure('Color','w');

contourf(X,Y,tau/1e6,30,'LineColor','none');
colormap("turbo")
hold on;
axis equal;
grid on;
colorbar;
xlabel('x (m)');
ylabel('y (m)');
title('Resultant Distribution (MPa)');

% PLOT Tau_xz SHEAR-STRESS CONTOUR
figure('Color','w');

contourf(X,Y,tau_xz/1e6,30,'LineColor','none');
colormap("turbo")
hold on;
axis equal;
grid on;
colorbar;
xlabel('x (m)');
ylabel('y (m)');
title('\tau_{xz} Distribution (MPa)');
% PLOT tau_yz SHEAR-STRESS CONTOUR
figure('Color','w');

contourf(X,Y,tau_yz/1e6,30,'LineColor','none');
colormap("turbo")
hold on;
axis equal;
grid on;
colorbar;
xlabel('x (m)');
ylabel('y (m)');
title('\tau_{yz} Distribution (MPa)');
fprintf('Resultant stress at centroid = %.6e Pa\n', min(min(tau)));
```

```
Resultant stress at centroid = 0.000000e+00 Pa
```

```
fprintf(' Maximum Resultant stress  = %.6e Pa\n', max(max(tau)));
```

```
 Maximum Resultant stress  = 5.000000e-01 Pa
```

Problem 2 : Open Thin walled section

```
T = 100;              % Applied torque (N.m)
G = 80e9;             % Shear modulus (Pa)
a = 0.20;             % Web length (m)
b = 0.10;             % Flange length (m)
t = 0.002;            % Thin-wall thickness (m)

alpha = 30*pi/180;    % Inclination of top strip

% TORSIONAL CONSTANT OF EACH STRIP
J_web = (1/3)*a*t^3;
J_top = (1/3)*b*t^3;
J_bottom = (1/3)*b*(2*t)^3;

J_total = J_web + J_top + J_bottom; K = T/(G*J_total);
tau_web = G*K*t;
tau_top = G*K*t;
tau_bottom = G*K*(2*t);  tau_max = max([tau_web,tau_top,tau_bottom]);

% creating the thin walled c/s
bottom = polyshape([0 b b 0],[0 0 2*t 2*t]);
web = polyshape([0 0 t t],[0 a a 0]);
figure(1);
A = union(bottom,web);
plot(A);
```

Problem 4

```

```

Elliptical c/s

```
clear all;
T=1000;  G=70e9;  % N and Pa
% major axis : a & minor axis: b
a=1; b=1*0.5;

% create a elliptical mesh grid
N = 501;

x = linspace(-a,a,N);
y = linspace(-b,b,N);

[X,Y] = meshgrid(x,y);
```

```
% Points inside the ellipse
inside = (X.^2/a^2 + Y.^2/b^2) <= 1;

% Torsional Constant
J = pi*(a*b)^3/(a^2+b^2);

% Prandtl's stress function
syms  phi x_ y_ a_ b_

phi = -T*((x_/a_)^2+(y_/b_)^2 -1)/(pi*a_*b_)
```

phi =

$$-\frac{\frac{1000\,x_^2}{a_^2}+\frac{1000\,y_^2}{b_^2}-1000}{a_\,b_\,\pi}$$

```
t_xz = diff(phi,y_)
```

t_xz =

$$-\frac{2000\,y_}{a_\,b_^3\,\pi}$$

```
t_yz = -diff(phi,x_)
```

t_yz =

$$\frac{2000\,x_}{a_^3\,b_\,\pi}$$

```
tau_xz = -2000*Y/(pi*a*b^3) ;  tau_yz = 2000*X/(pi*a^3*b^1);
tau = sqrt(tau_xz.^2 + tau_yz.^2);

Phi = (T/(pi*a*b)).*(1 - X.^2/a^2 - Y.^2/b^2);
% Remove values outside the ellipse
tau_xz(~inside) = NaN;
tau_yz(~inside) = NaN;
tau(~inside) = NaN;
Phi(~inside) =NaN;



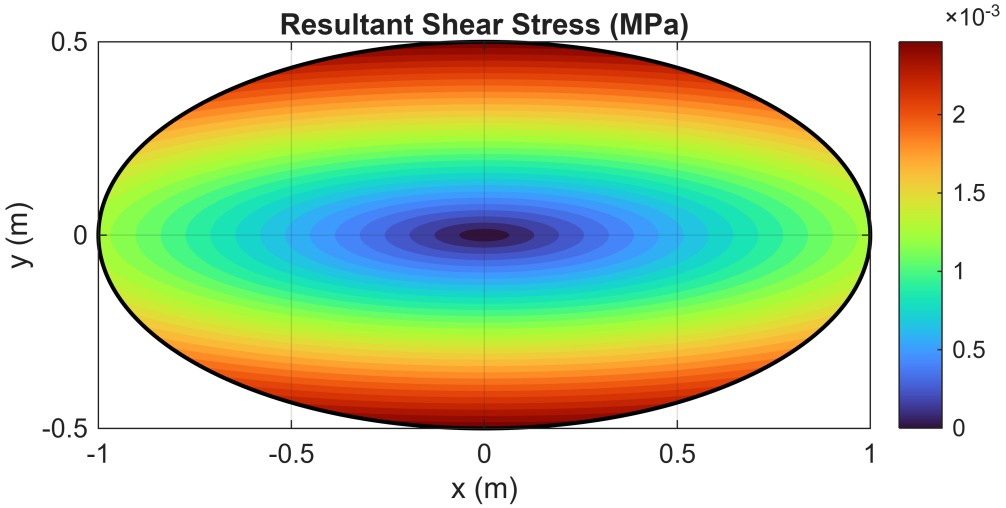
% PLOT RESULTANT SHEAR-STRESS CONTOUR
figure('Color','w');

contourf(X,Y,tau/1e6,30,'LineColor','none');
colormap("turbo")
hold on;

plot(a*cos(linspace(0,2*pi,1000)), ...
     b*sin(linspace(0,2*pi,1000)), ...
     'k','LineWidth',1.5);
```

```
axis equal;
grid on;
colorbar;

xlabel('x (m)');
ylabel('y (m)');
title('Resultant Shear Stress (MPa)');
```



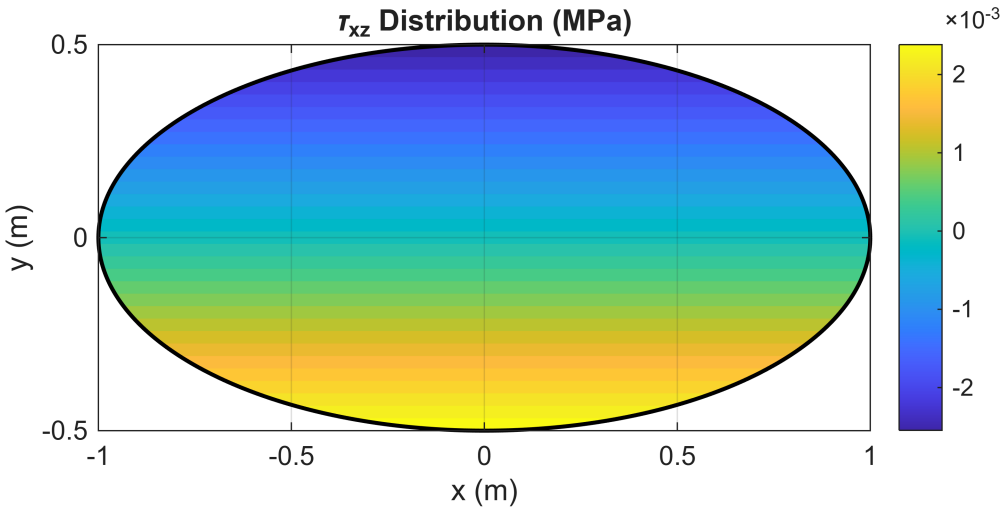

```
%% 10. PLOT tau_xz DISTRIBUTION
figure('Color','w');

contourf(X,Y,tau_xz/1e6,30,'LineColor','none');
hold on;

plot(a*cos(linspace(0,2*pi,500)), ...
     b*sin(linspace(0,2*pi,500)), ...
     'k','LineWidth',1.5);

axis equal;
grid on;
colorbar;

xlabel('x (m)');
ylabel('y (m)');
title('\tau_{xz} Distribution (MPa)');
```



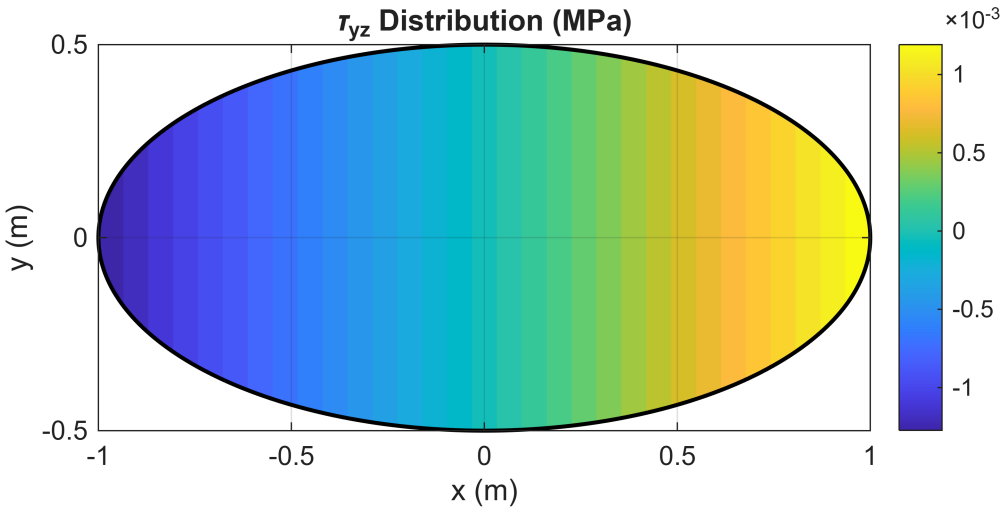

```
%% 11. PLOT tau_yz DISTRIBUTION
figure('Color','w');

contourf(X,Y,tau_yz/1e6,30,'LineColor','none');
hold on;

plot(a*cos(linspace(0,2*pi,500)), ...
     b*sin(linspace(0,2*pi,500)), ...
     'k','LineWidth',1.5);

axis equal;
grid on;
colorbar;

xlabel('x (m)');
ylabel('y (m)');
title('\tau_{yz} Distribution (MPa)');
```




```
% PLOT SHEAR-STRESS VECTORS
% Subsample the grid to keep the arrows readable

figure('Color','w');

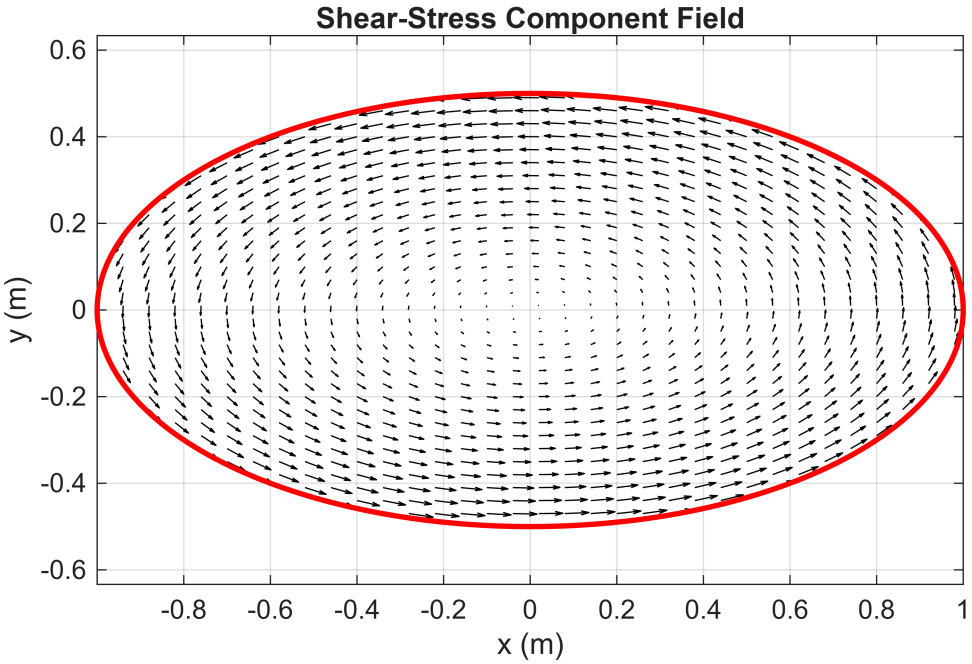
skip = 15;
Xs = X(1:skip:end,1:skip:end);
Ys = Y(1:skip:end,1:skip:end);

Tx = tau_xz(1:skip:end,1:skip:end);
Ty = tau_yz(1:skip:end,1:skip:end);

quiver(Xs,Ys,Tx,Ty,'k');
hold on;

plot(a*cos(linspace(0,2*pi,500)), ...
     b*sin(linspace(0,2*pi,500)), ...
     'r','LineWidth',2);

axis equal;
grid on;
xlabel('x (m)');
ylabel('y (m)');
title('Shear-Stress Component Field');
```



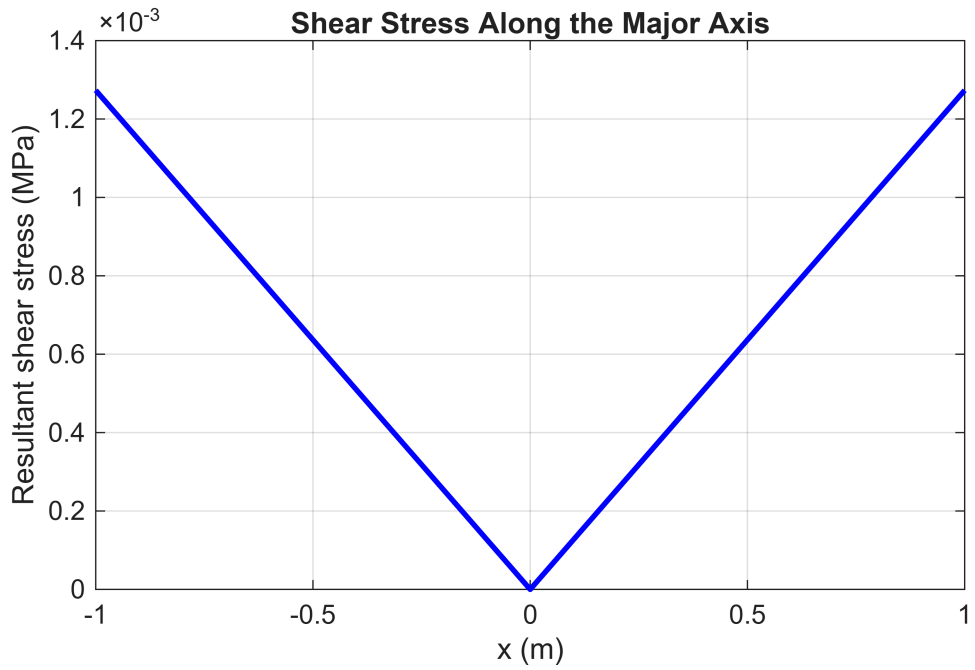

```
% STRESS ALONG THE MAJOR AXIS
x_axis = linspace(-a,a,501);
y_axis = zeros(size(x_axis));

tau_axis = sqrt( ...
    (-2*T*y_axis/(pi*a*b^3)).^2 + ...
    ( 2*T*x_axis/(pi*a^3*b)).^2 );

figure('Color','w');

plot(x_axis,tau_axis/1e6,'b','LineWidth',2);
hold on;
grid on;

xlabel('x (m)');
ylabel('Resultant shear stress (MPa)');
title('Shear Stress Along the Major Axis');
```




```
% STRESS ALONG THE Minor AXIS

y_axis = linspace(-b,b,501);
x_axis = zeros(size(y_axis));

tau_axis = sqrt( ...
    (-2*T*y_axis/(pi*a*b^3)).^2 + ...
    ( 2*T*x_axis/(pi*a^3*b)).^2 );

figure('Color','w');

plot(y_axis,tau_axis/1e6,'r','LineWidth',2);
grid on;

xlabel('x (m)');
ylabel('Resultant shear stress (MPa)');
title('Shear Stress Along the Minor Axis');
hold off;
```




```
%% 14. VERIFY MAXIMUM NUMERICALLY
fprintf('Numerical maximum on grid = %.6f MPa\n', ...
        max(tau(:))/1e6);
```

```
Numerical maximum on grid = 0.002546 MPa
```

Warping Phenomenon

```
% In notes we get
W  = (T*(b^2-a^2)/(G*pi*a^3*b^3)).*(X.*Y);
W(~inside)=NaN;
figure('Color','w');

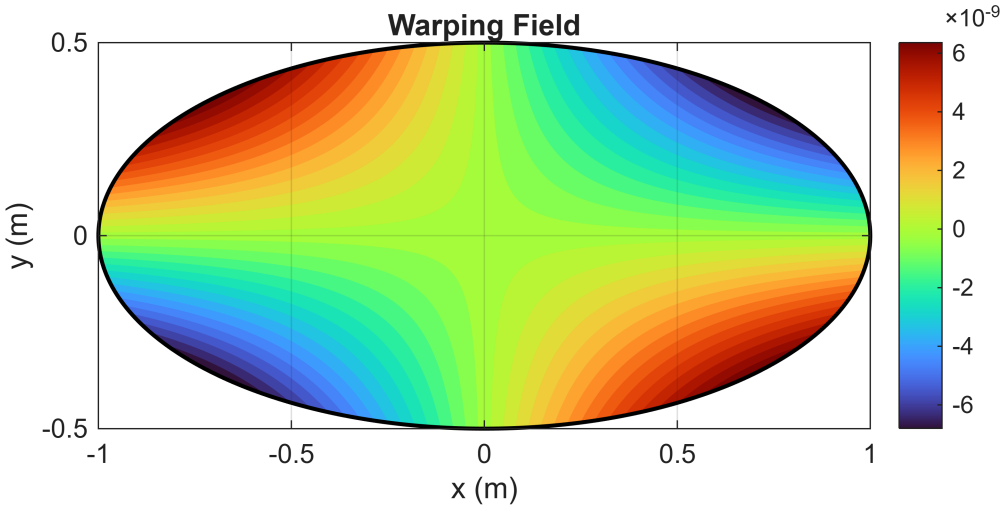
contourf(X,Y,W,30,'LineColor','none');
colormap("turbo")
hold on;

plot(a*cos(linspace(0,2*pi,1000)), ...
    b*sin(linspace(0,2*pi,1000)), ...
    'k','LineWidth',1.5);

axis equal;
grid on;
colorbar;

xlabel('x (m)');
ylabel('y (m)');
```

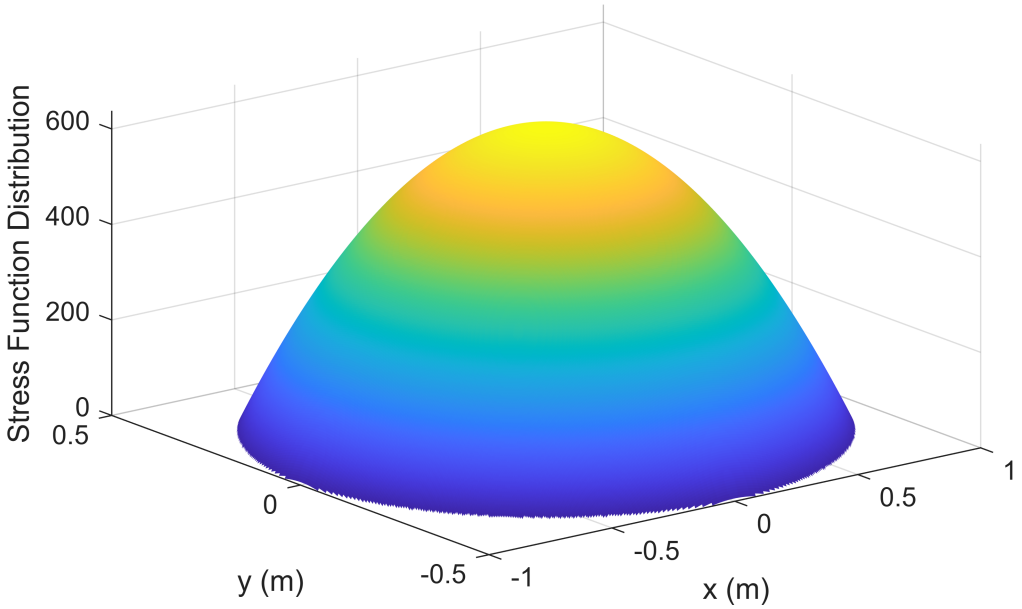
```
title('Warping Field');
```



3D plot

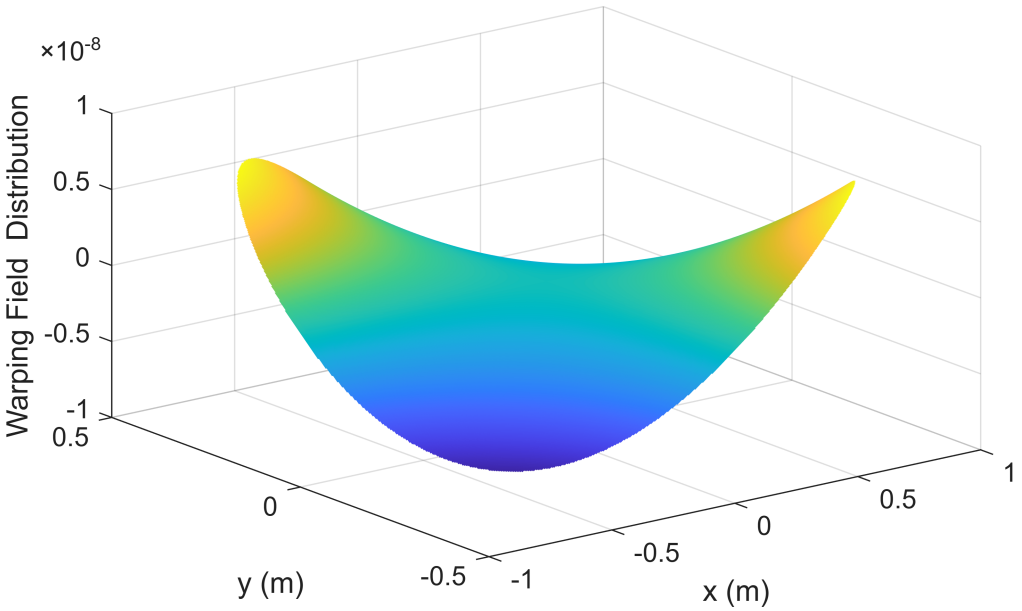
```
%% 3D plot
figure('Color','w');

surf(X,Y,Phi,'EdgeColor','none');
hold on;
xlabel('x (m)');
ylabel('y (m)');
zlabel('Stress Function Distribution');
```



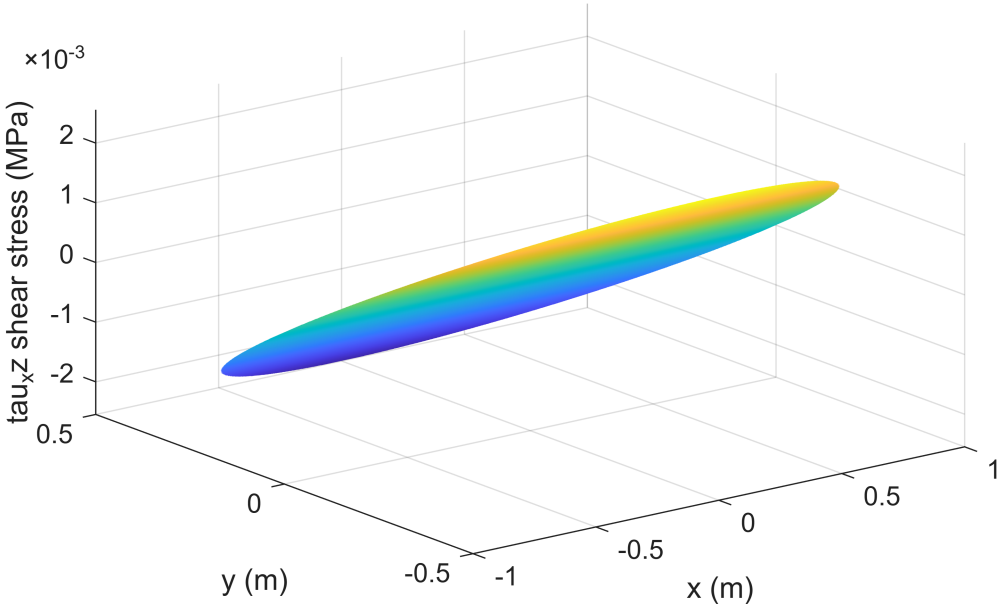

```
%% 3D plot
figure('Color','w');

surf(X,Y,W,'EdgeColor','none');
hold on;
xlabel('x (m)');
ylabel('y (m)');
zlabel('Warping Field  Distribution');
```



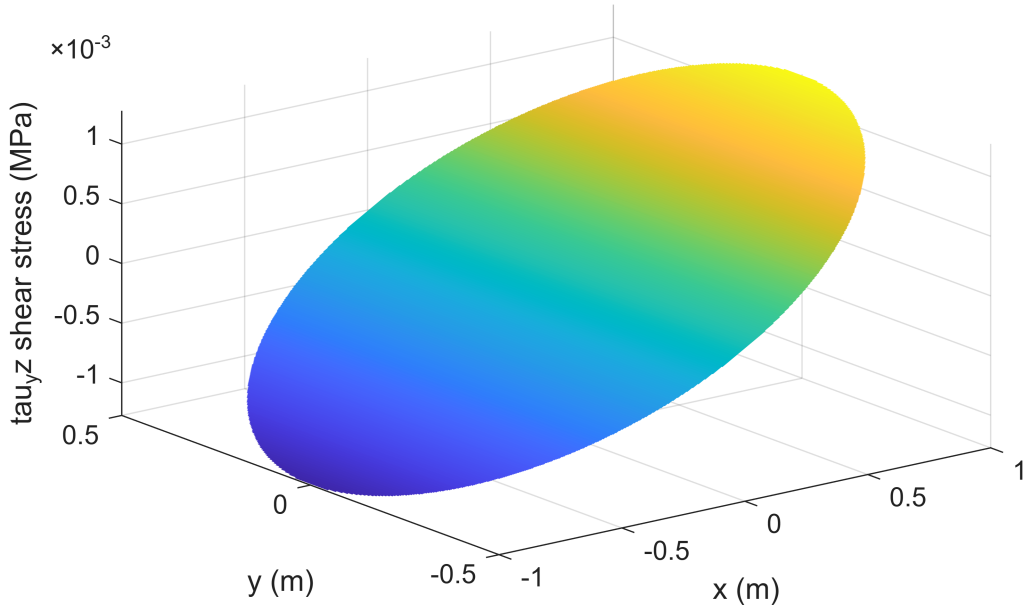

```
figure('Color','w');
```

```
surf(X, Y, tau_xz/1e6, tau_xz/1e6, ...
    'EdgeColor','none');
xlabel('x (m)');
ylabel('y (m)');
zlabel('tau_xz shear stress (MPa)');
```




```
figure('Color','w');

surf(X, Y, tau_yz/1e6, tau_yz/1e6, ...
    'EdgeColor','none');
xlabel('x (m)');
ylabel('y (m)');
zlabel('tau_yz shear stress (MPa)');
```




```
figure('Color','w');

surf(X, Y, tau, tau, ...
    'EdgeColor','none');

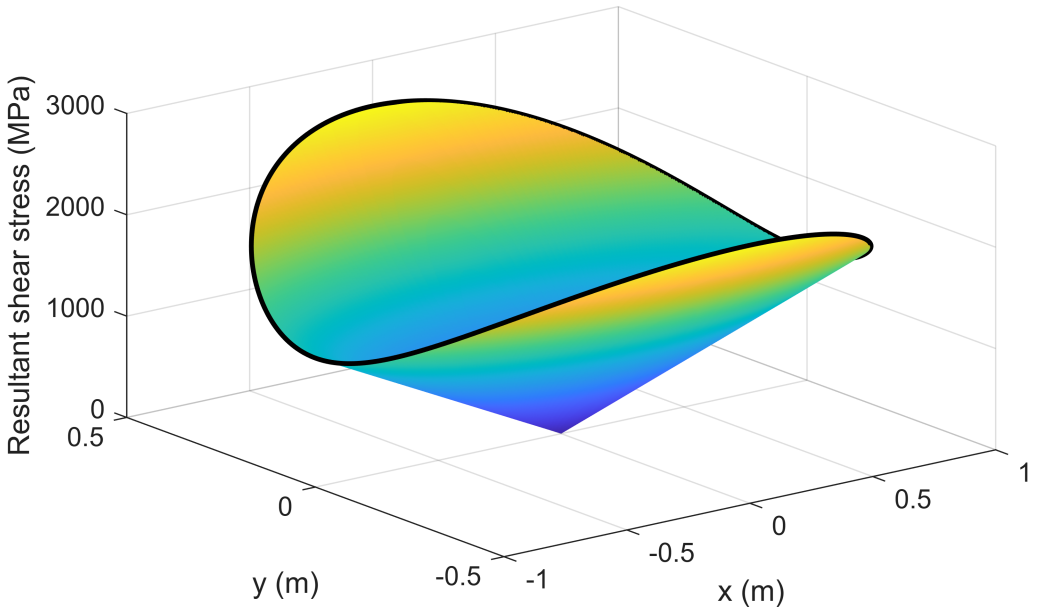
hold on;
theta = linspace(0,2*pi,500);

xb = a*cos(theta);
yb = b*sin(theta);

tau_boundary = sqrt( ...
    (2*T*yb/(pi*a*b^3)).^2 + ...
    (2*T*xb/(pi*a^3*b)).^2 );

plot3(xb, yb, tau_boundary, ...
    'k-', 'LineWidth', 1.8);

xlabel('x (m)');
ylabel('y (m)');
zlabel('Resultant shear stress (MPa)');
```


Resultant shear stress (MPa)
3000
2000
1000
0
0.5
0
-0.5
y (m)
-1
-0.5
0
0.5
1
x (m)